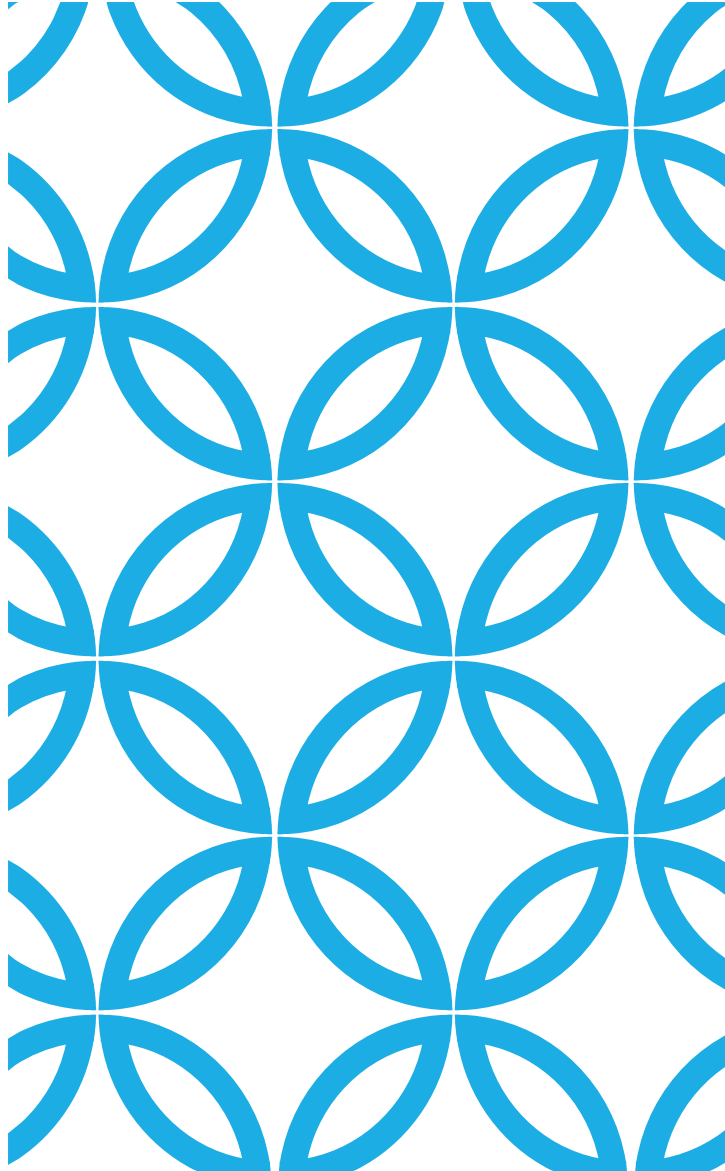




RESILIENCE

Dr Megan Boston
ENGEV370

LEARNING OBJECTIVES

01

Define Resilience

02

Relate resilience
curves to system
functionality

03

Quantify resilience

RESILIENCE

What is Resilience?

- How does Resilience relate to Sustainability?

DEFINITIONS

Risk – Risk Triplet

1. What can go wrong? (scenarios combining hazards and outcomes)
2. How likely is it to go wrong? (probability of the scenario)
3. What are the consequences if it goes wrong?

Vulnerability – the diminished capacity of an individual or group to anticipate, cope with, resist and recover from the impact of a natural or man-made hazard

Resilience – The capacity to absorb or mitigate the impact of hazard events while maintaining and restoring essential services

WELLINGTON NEW ZEALAND



DIMENSIONS OF RESILIENCE

Technical: The ability of physical systems to perform to acceptable or desired levels

Organizational: The capacity of organizations to make decisions and take actions that contribute to resilience

Social: Measures specifically designed to lessen the extent to which disaster stricken communities and governmental jurisdictions suffer negative consequence due to loss of critical services

Economical: The capacity to reduce both direct and indirect economic losses resulting from disaster

DIMENSIONS OF RESILIENCE

Technical

- The ability of physical systems to perform to acceptable or desired levels

Organizational

- The capacity of organizations to make decisions and take actions that contribute to resilience

Social

- Measures specifically designed to lessen the extent to which disaster-stricken communities and governmental jurisdictions suffer negative consequence due to loss of critical services

Economical

- The capacity to reduce both direct and indirect economic losses resulting from disaster

RESILIENCE CASE STUDIES

Questions to consider:

1. What are the short, medium, and long term impacts of this disaster on the community?
2. What can be learned to improve community resilience for future disasters?
3. What are the technical, social, organizational, and economic impacts of these disasters?



Great Hanshin (“Kobe”) Earthquake of 1995

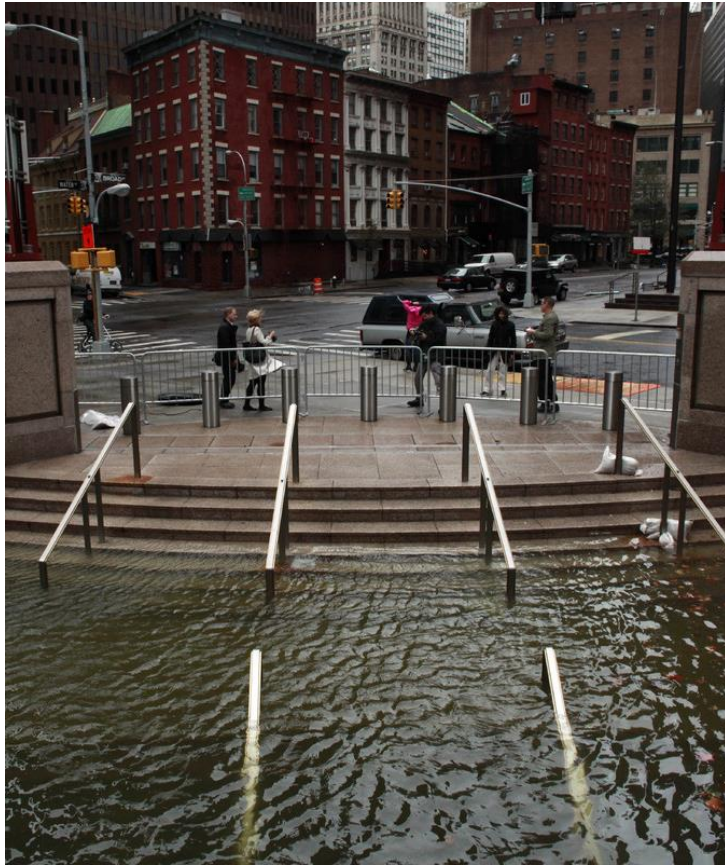
- 6,434 Fatalities
- 400,000 Buildings Destroyed
- 300 Fires



Northeastern Blackout of 2003

- 55 Million people affected
- 8 US States and parts of Canada
- 1 day to 1 week to restore power
- Many infrastructure systems disrupted

<https://www.youtube.com/watch?v=nd3teNgUq8E>



Hurricane Sandy, 29 October 2012

- \$75 Billion in Damage
- 233 Fatalities
- 15 states

Subways
Roads
EMT/Fire
Hospitals
Power
Airports

Water/wastewater
Data centers
Homes (displacement)
Gas rationing



CHRISTCHURCH 2011



Christchurch Earthquake (magnitude 6.2)

22 February 2011

185 Fatalities

Over 100,000 houses damaged



Flooding around Gore. Photo: High Country Hslicopster



Milford Rd State Highway 94. Photo: NZTA



Lake Howden Hut, Photo: DOC

SOUTHLAND FLOOD 2020



Milford Rd State Highway 94. Photo: NZTA

SOUTHLAND FLOODS 2020

1100 mm of rain over 60 hr period

~200 tourists in Milford Sound

~200 staff in Milford Sound

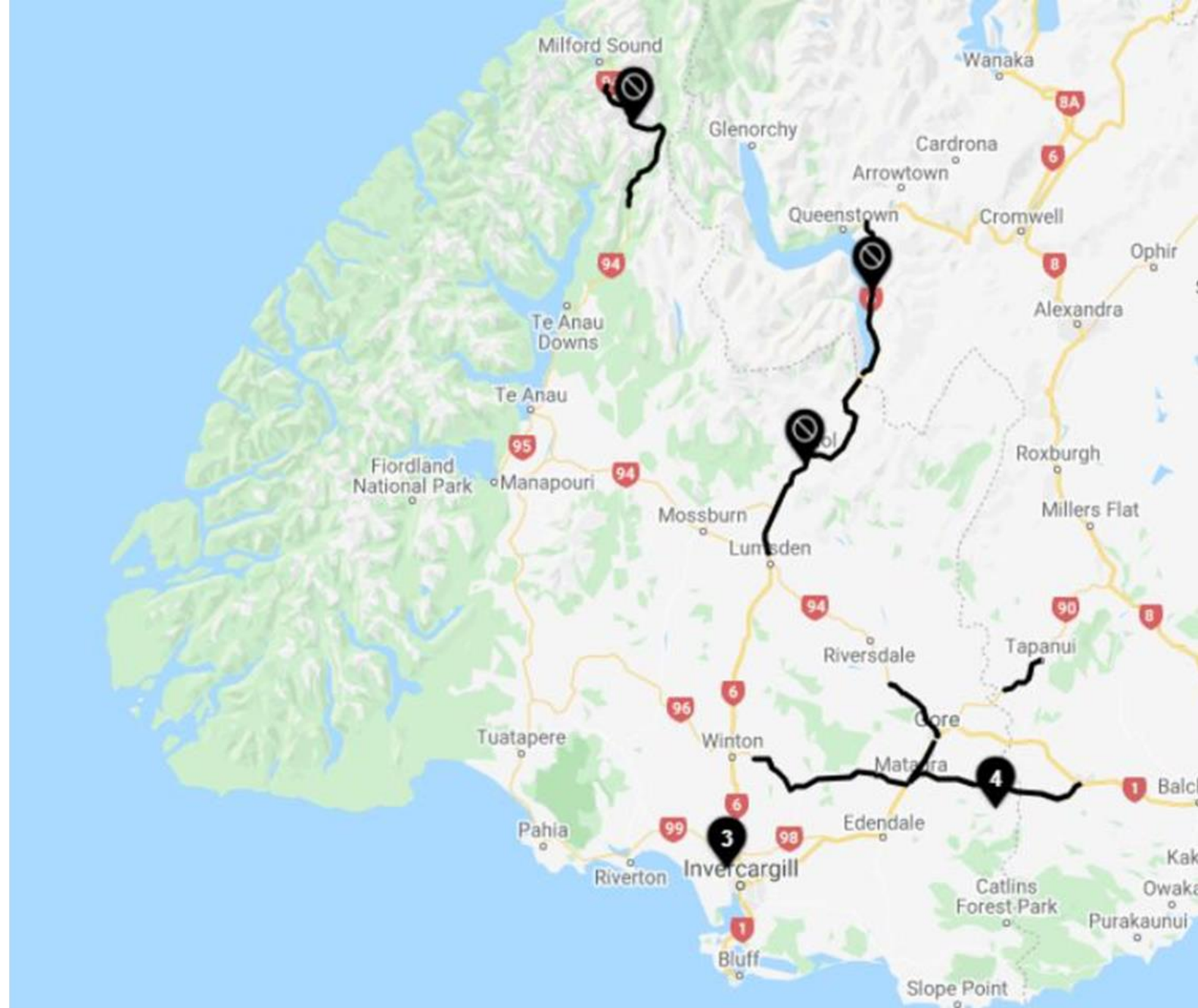
~70 people stranded along Milford road

100+ people trapped in Great Walk and backcountry Huts

6000 people evacuated from Gore, Wyndham, and Mataura

Milford Road closed for weeks, partial reopening to convoys only

SOUTHLAND FLOODS 2020



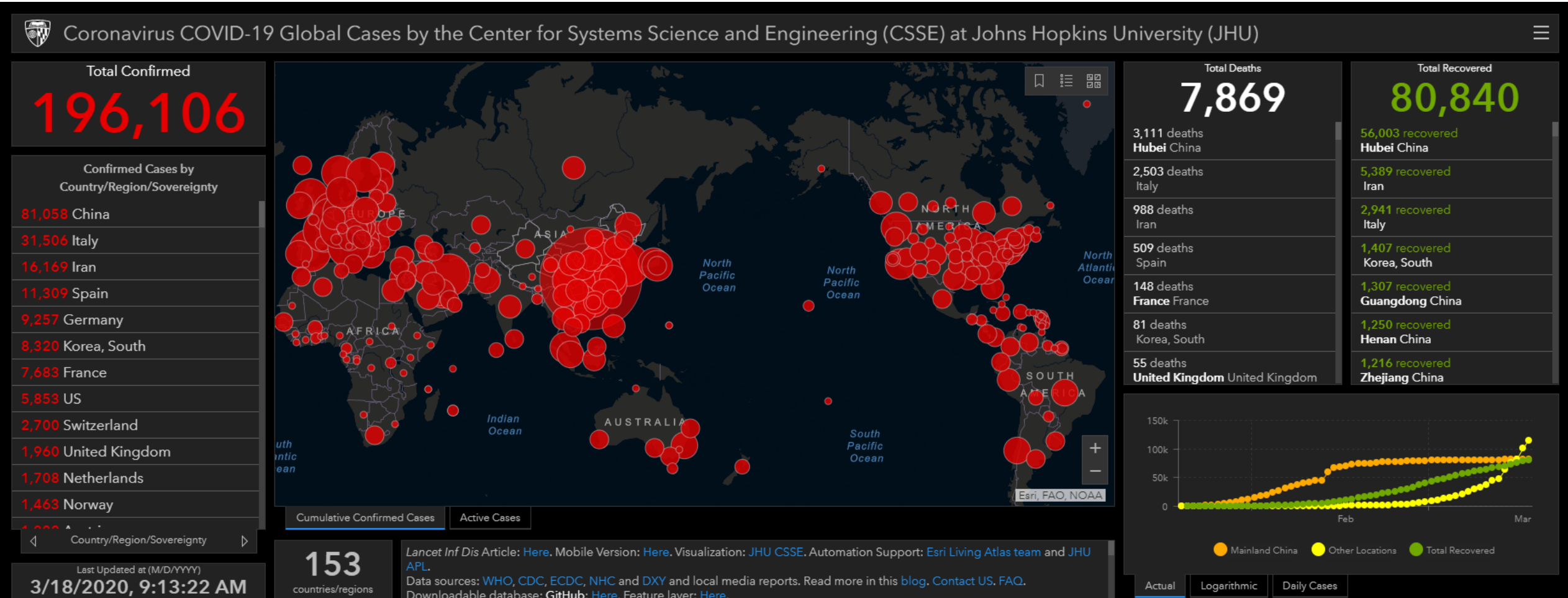
Road Closures due to Southland Flooding, Photo:NZTA



COVID-19

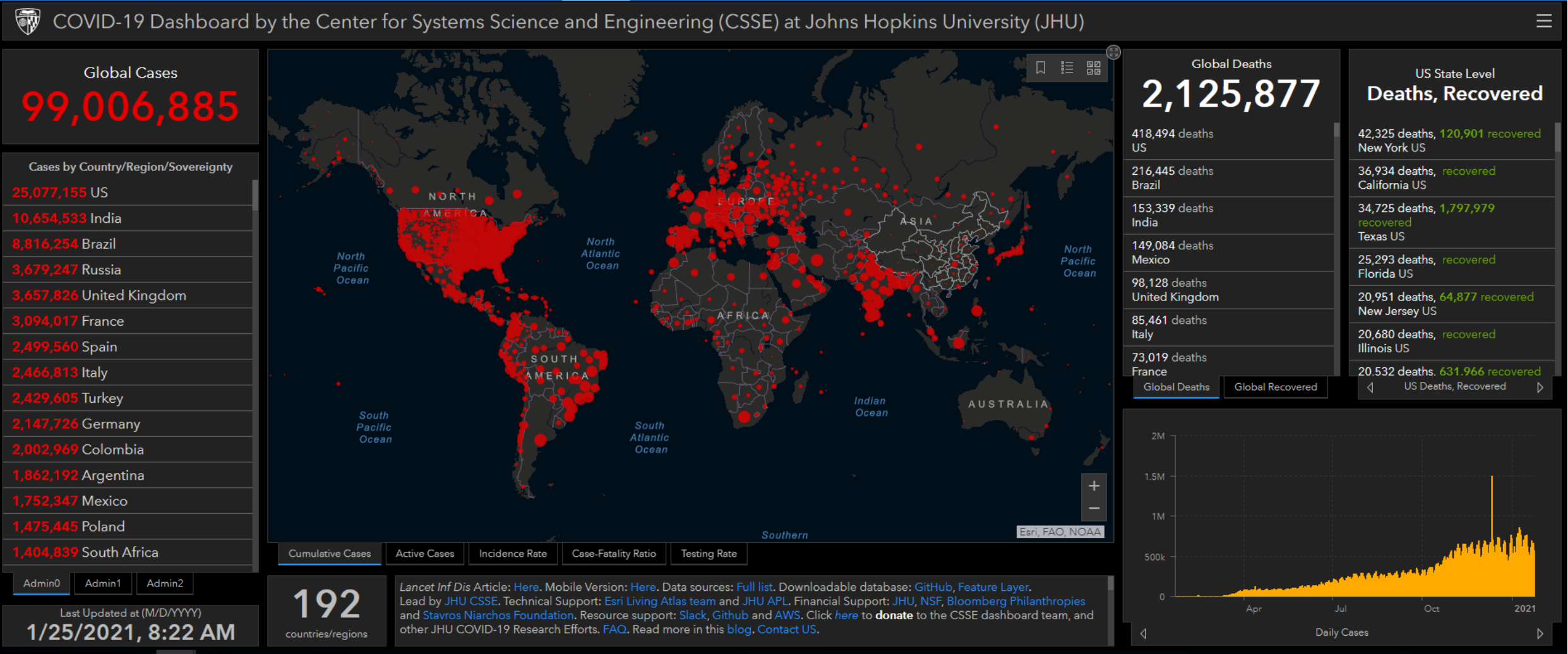
Travel Restrictions
Social Distancing

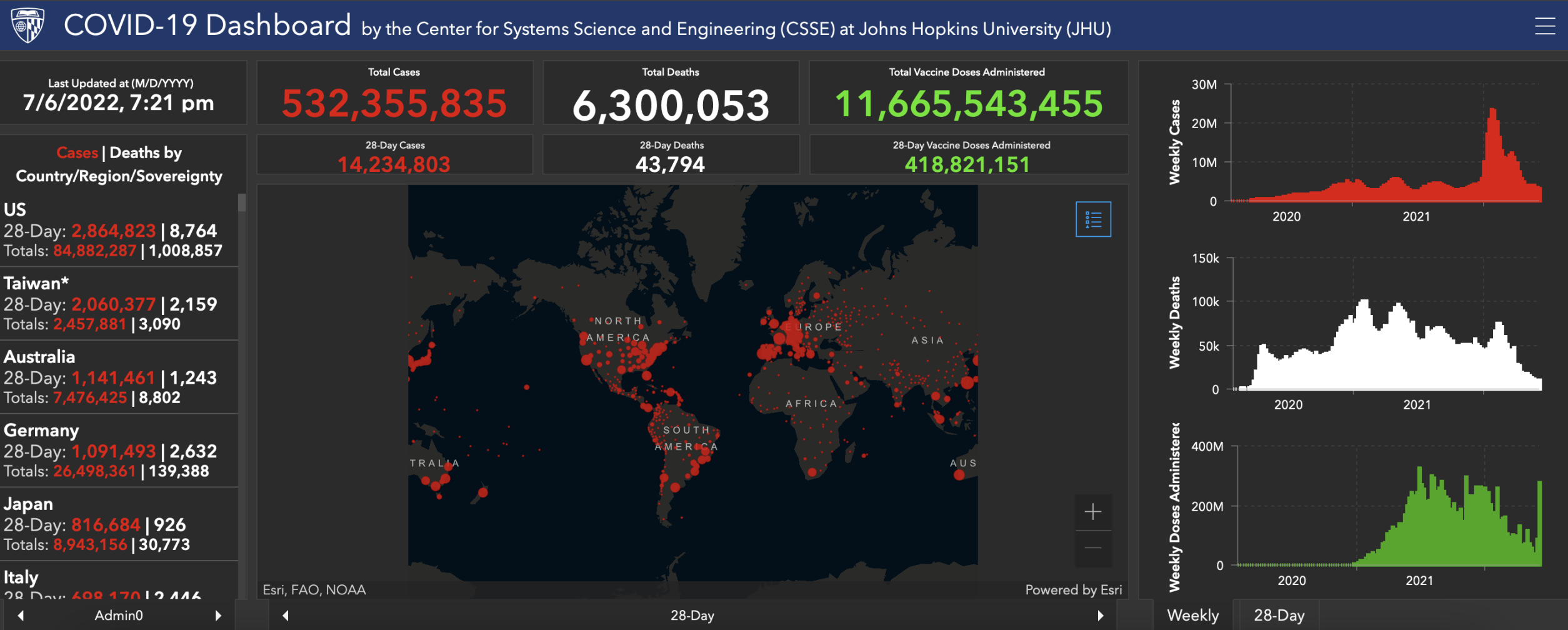
2020



Travel Restrictions
Social Distancing

January 2021





IN SUMMARY - NATURAL (AND UNNATURAL) DISASTERS:

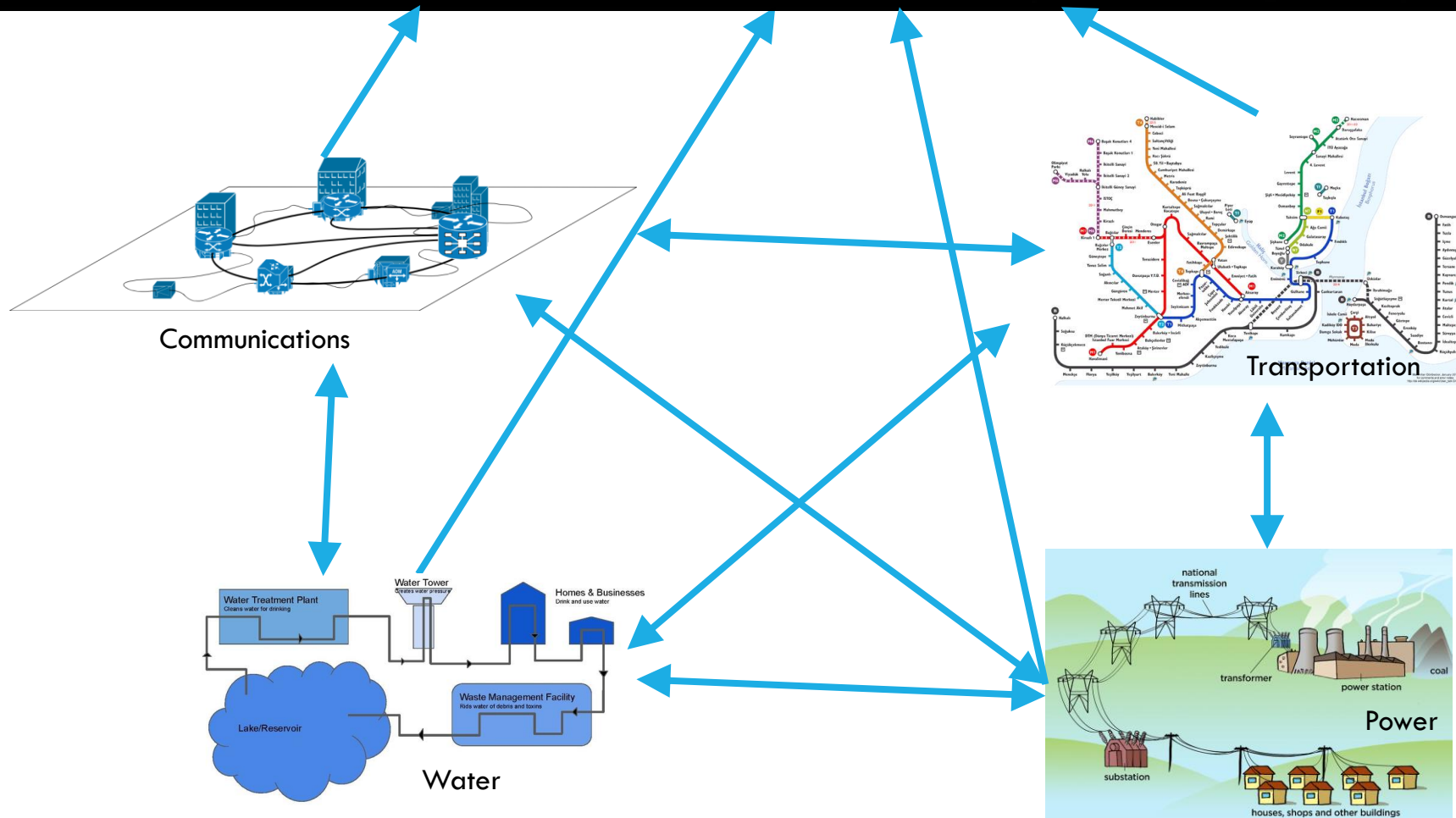
1. Widespread impacts across multiple sectors of infrastructure and society
2. Emergency response efforts often impeded by infrastructure failures
3. Cannot simply think in terms of infrastructure components, but must consider system aspects
4. How do you decide the levels of risk you are willing to tolerate?

RESILIENCE

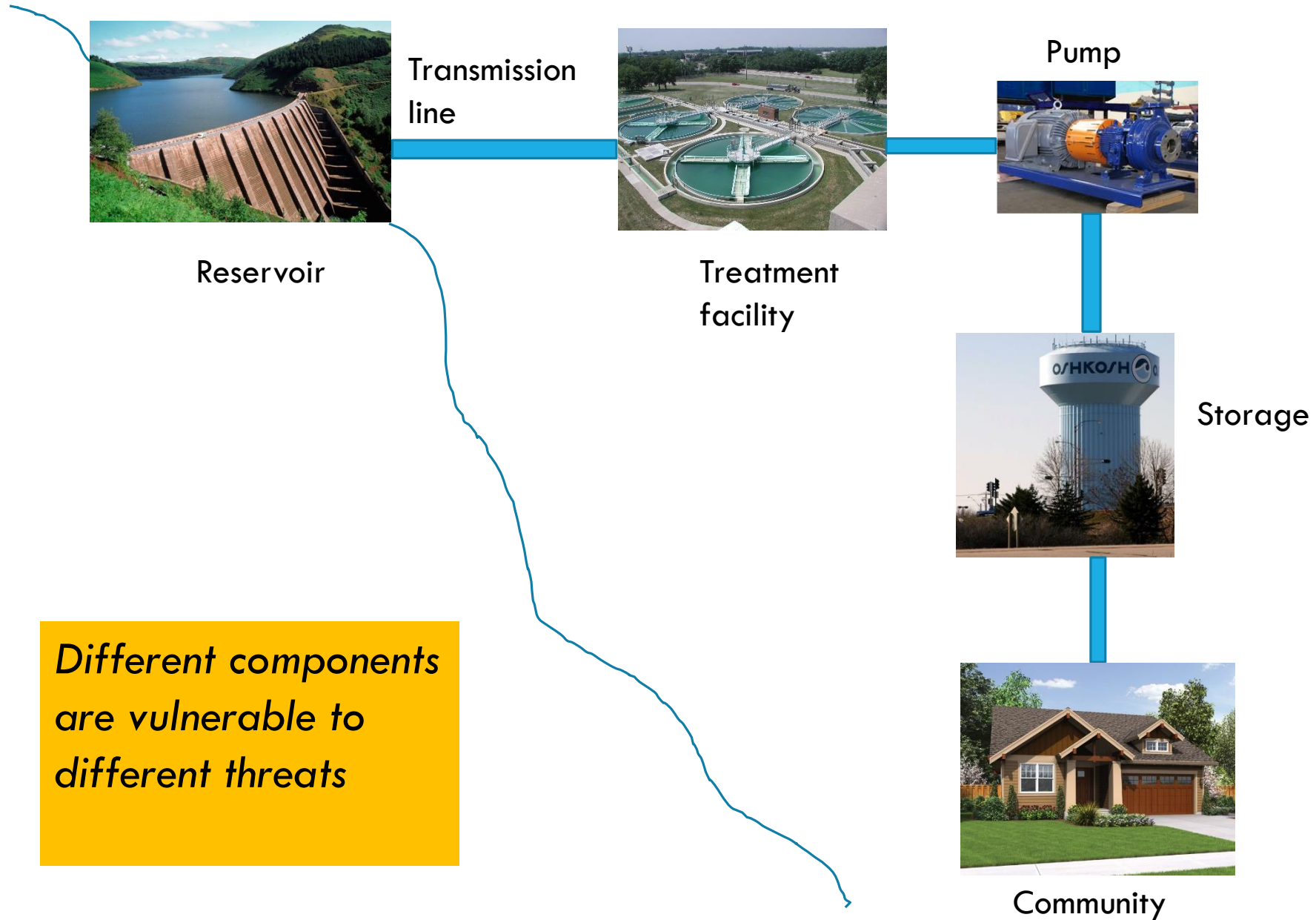
The ability to either withstand a hazard event or recover quickly with minimal disruption of service.

Resilience is inherently a systems aspect of infrastructure because delivering an infrastructure service requires interaction of many components and sectors.

How do you design for resilience?

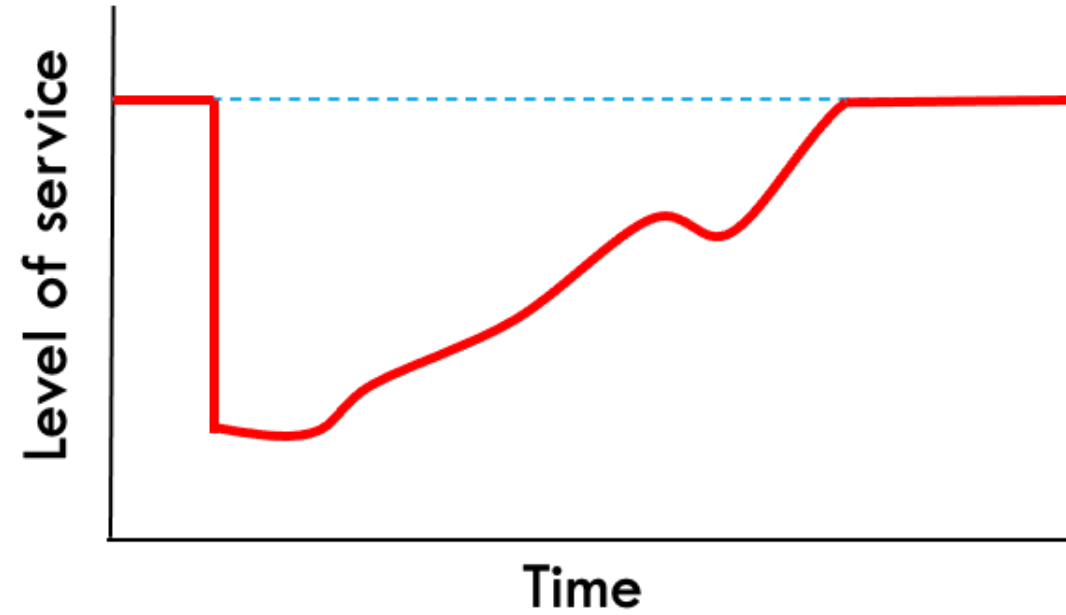


Water Supply Example



The 4 R's of Resilience



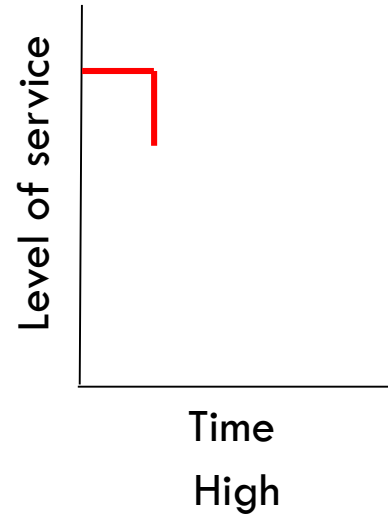
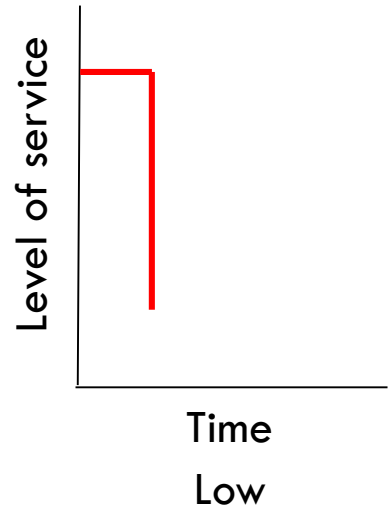


From Bruneau et al. (2003)

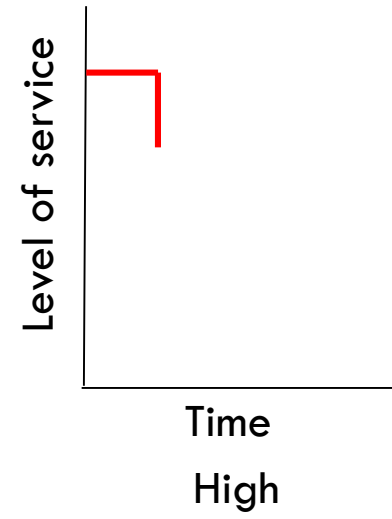
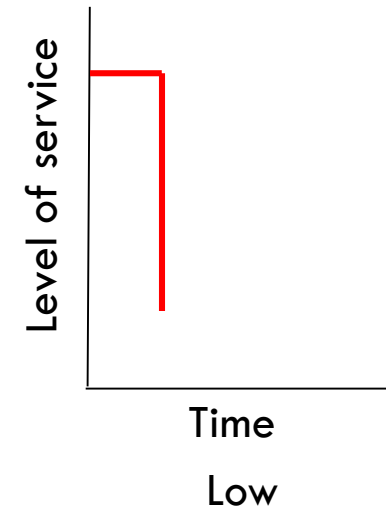


RESILIENCE TRIANGLE

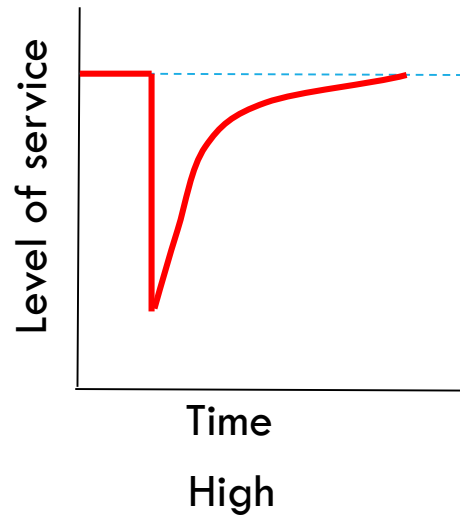
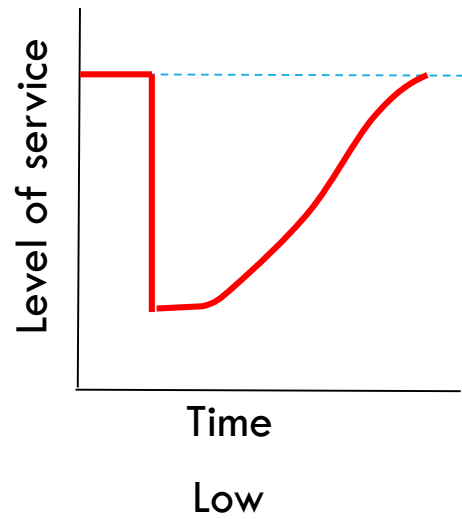
Robustness



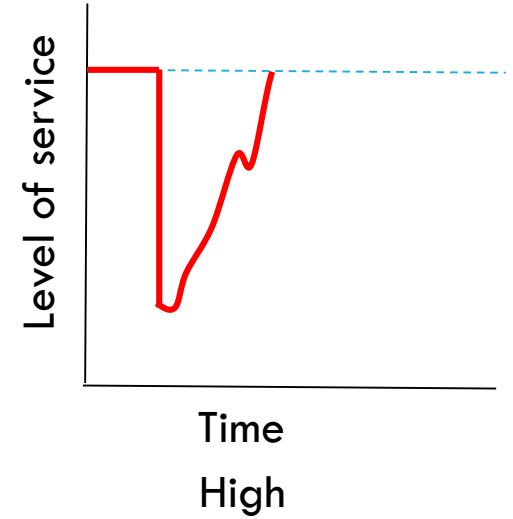
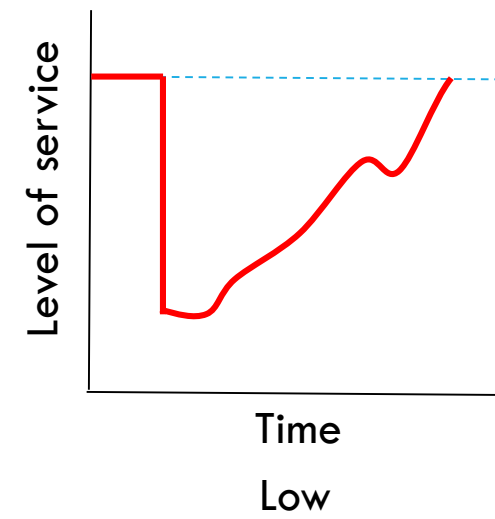
Redundancies



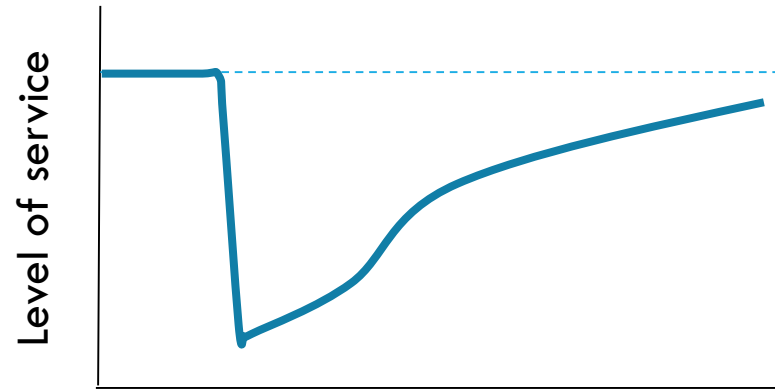
Resourcefulness



Rapidity

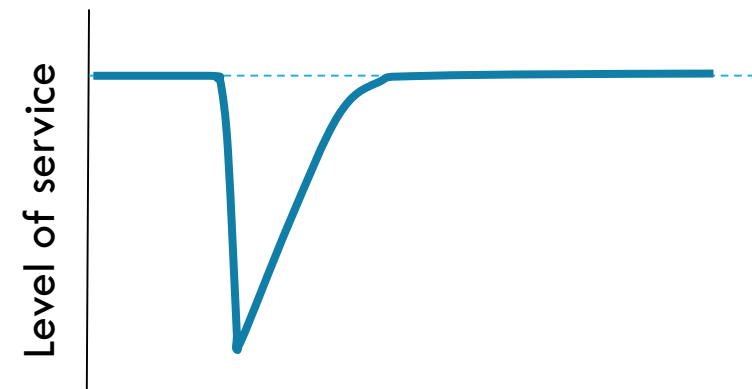


Response of four communities to the same disruption



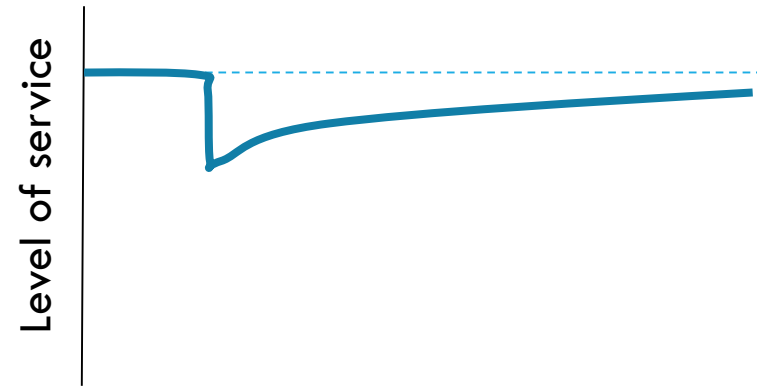
Time

Robustness	☐ High	☒ Low
Redundancy	☐ High	☒ Low
Rapidity	☐ High	☒ Low
Resourcefulness	☐ High	☒ Low



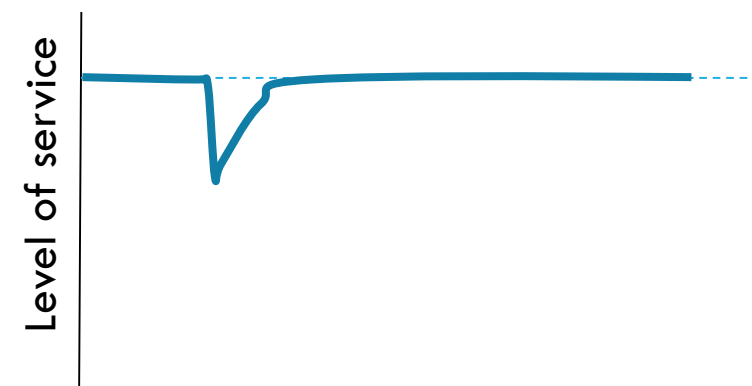
Time

Robustness	☐ High	☒ Low
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Rapidity	☒ High	☐ Low
Resourcefulness	☒ High	☐ Low



Time

Robustness	☒ High	☐ Low
Redundancy	☒ High	☐ Low
Rapidity	☐ High	☒ Low
Resourcefulness	☐ High	☒ Low



Time

Robustness	☒ High	☐ Low
Redundancy	☒ High	☐ Low
Rapidity	☒ High	☐ Low
Resourcefulness	☒ High	☐ Low

RESILIENCE PRACTICE

For your project topic:

- What are the potential hazards?
- What are the risks associated with the project?

How does your project interface with the 4 dimension of resilience?

- Technological
- Organizational
- Social
- Economical

How can you use the 4 properties of resilience to decrease the risk?

- Robustness
- Redundancy
- Resourcefulness
- Rapidity



**WHAT DOES RESILIENCE
HAVE TO DO WITH
SUSTAINABILITY?**